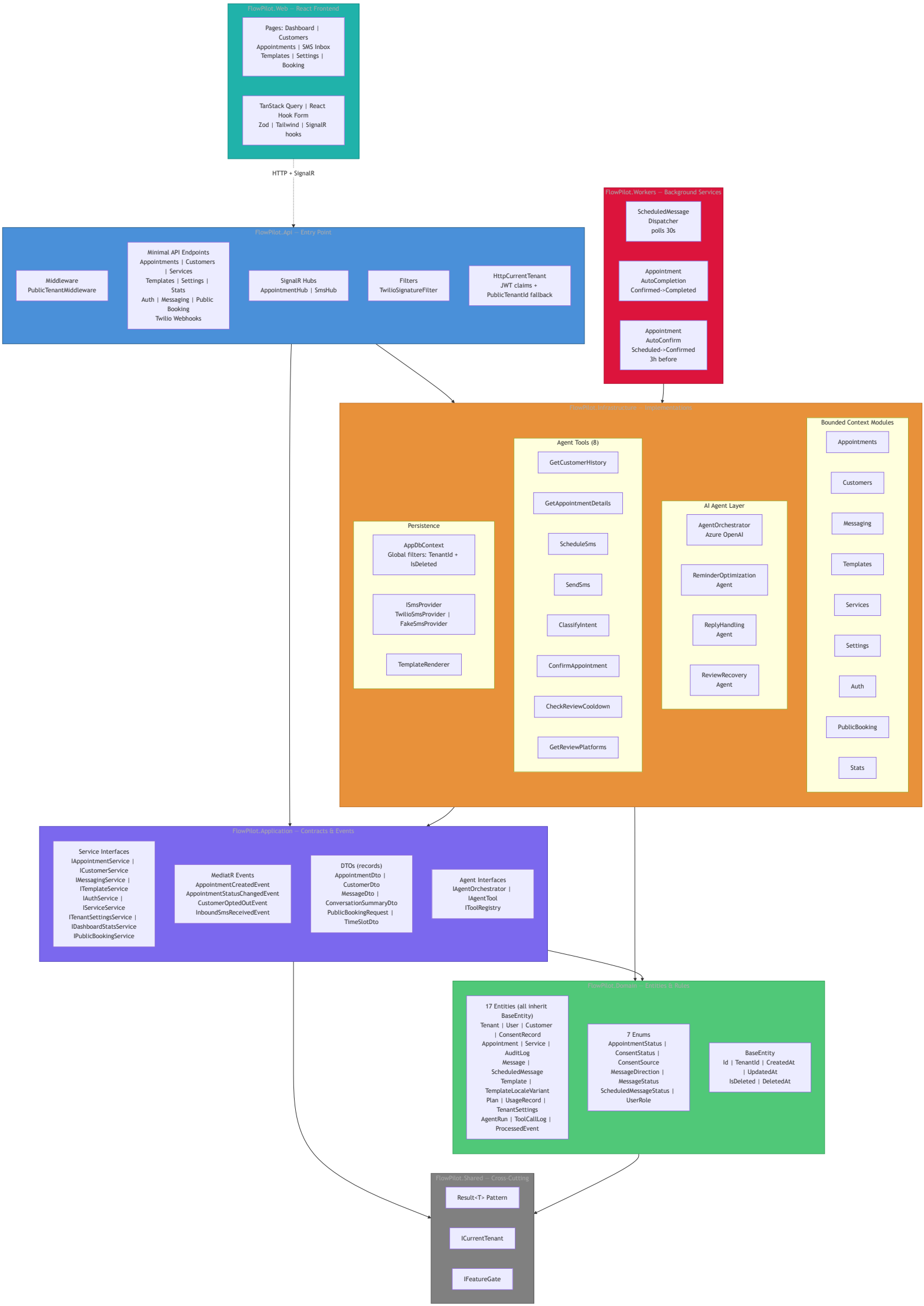
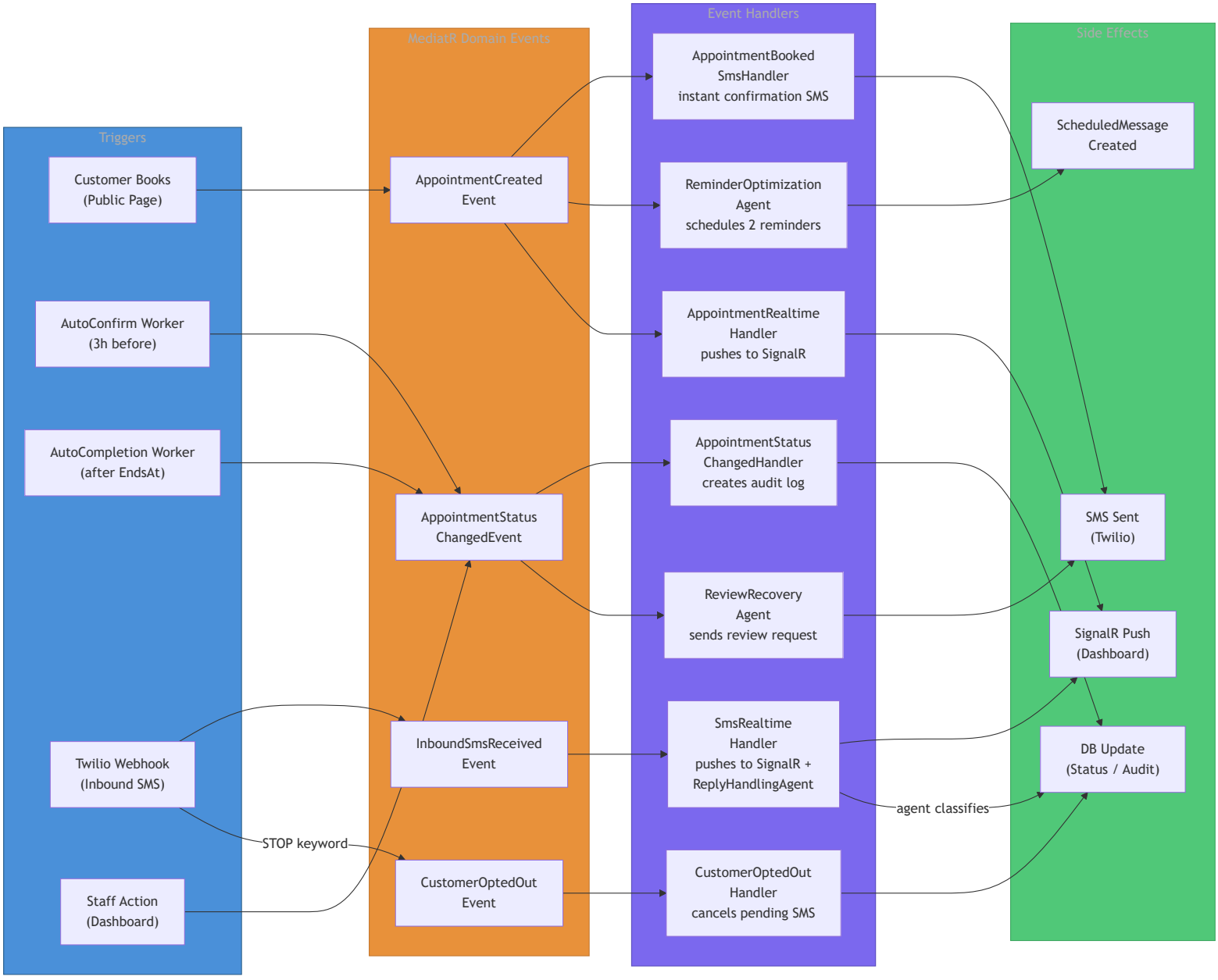
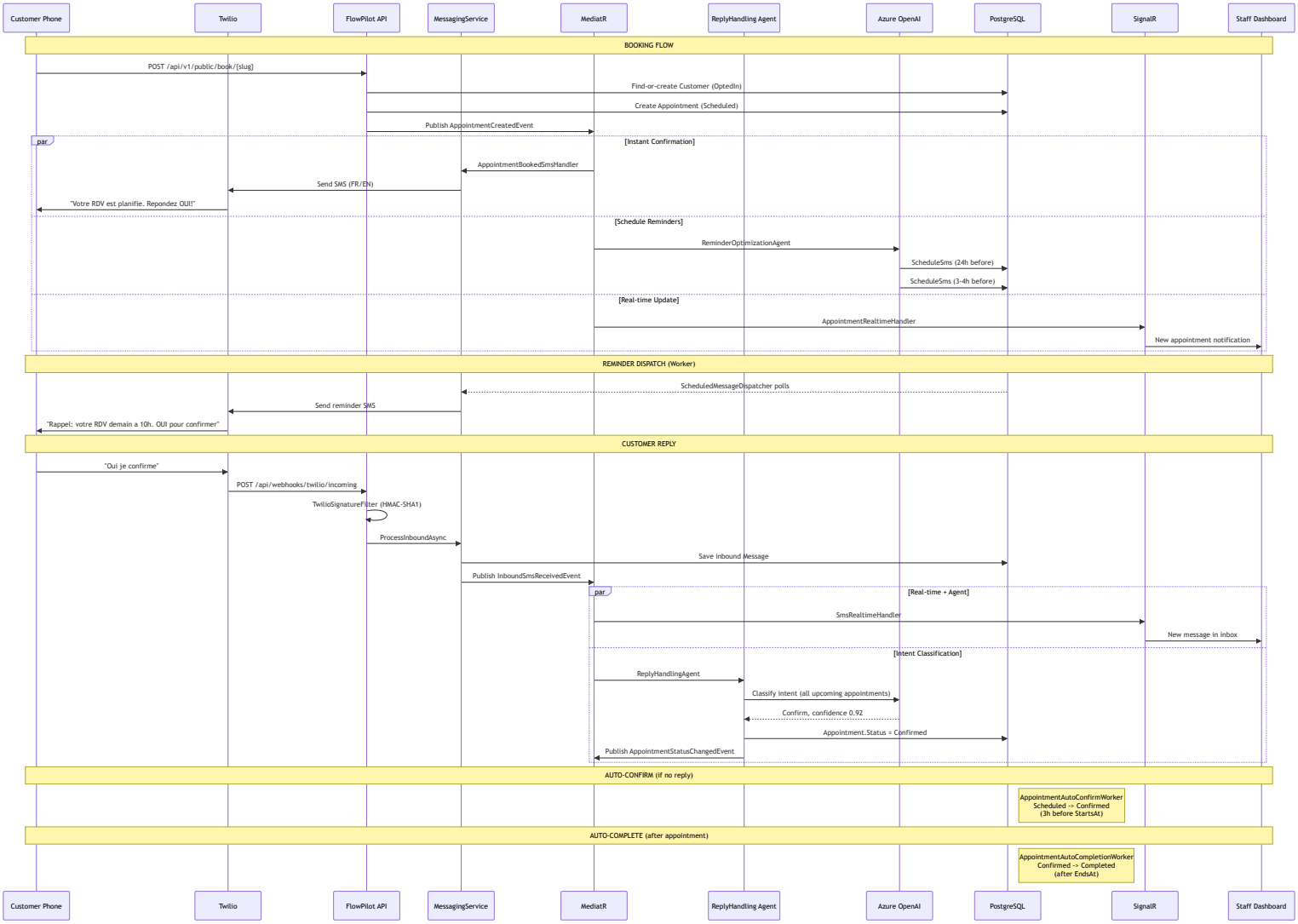
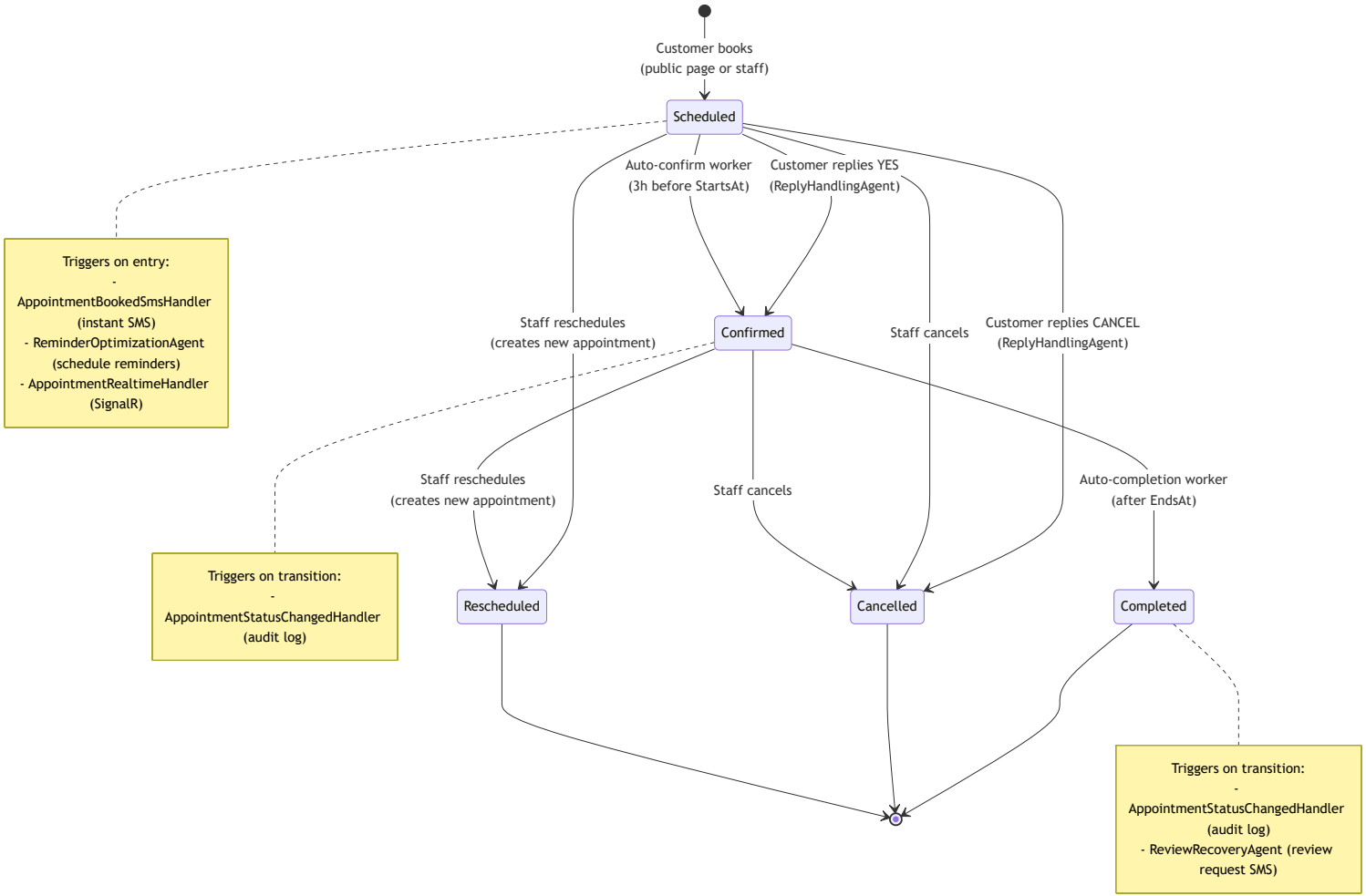

FlowPilot.Web — React Frontend
Pages: Dashboard | Customers Appointments | SMS Inbox Templates | Settings | Booking
TanStack Query | React Hook Form Zod | Tailwind | SignalR hooks
HTTP + SignalR
FlowPilot.Workers — Background Services
ScheduledMessage Dispatcher polls 30s
Appointment AutoCompletion Confirmed->Completed
Appointment AutoConfirm Scheduled->Confirmed 3h before
FlowPilot.Api — Entry Point
Middleware PublicTenantMiddleware
Minimal API Endpoints Appointments | Customers | Services Templates | Settings | Stats Auth | Messaging | Public Booking Twilio Webhooks
SignalR Hubs AppointmentHub | SmsHub
Filters TwilioSignatureFilter
HttpCurrentTenant JWT claims + PublicTenantId fallback
FlowPilot.Infrastructure — Implementations
Persistence
AppDbContext Global filters: TenantId + IsDeleted
ISmsProvider TwilioSmsProvider | FakeSmsProvider
TemplateRenderer
Agent Tools (8)
GetCustomerHistory
GetAppointmentDetails
ScheduleSms
SendSms
ClassifyIntent
ConfirmAppointment
CheckReviewCooldown
GetReviewPlatforms
AI Agent Layer
AgentOrchestrator Azure OpenAI
ReminderOptimization Agent
ReplyHandling Agent
ReviewRecovery Agent
Bounded Context Modules
Appointments
Customers
Messaging
Templates
Services
Settings
Auth
PublicBooking
Stats
FlowPilot.Application — Contracts & Events
Service Interfaces IAppointmentService | ICustomerService IMessagingService | ITemplateService IAuthService | IServiceService ITenantSettingsService | IDashboardStatsService IPublicBookingService
MediatR Events AppointmentCreatedEvent AppointmentStatusChangedEvent CustomerOptedOutEvent InboundSmsReceivedEvent
DTOs (records) AppointmentDto | CustomerDto MessageDto | ConversationSummaryDto PublicBookingRequest | TimeSlotDto
Agent Interfaces IAgentOrchestrator | IAgentTool IToolRegistry
FlowPilot.Domain — Entities & Rules
17 Entities (all inherit BaseEntity) Tenant | User | Customer | ConsentRecord Appointment | Service | AuditLog Message | ScheduledMessage Template | TemplateLocaleVariant Plan | UsageRecord | TenantSettings AgentRun | ToolCallLog | ProcessedEvent
7 Enums AppointmentStatus | ConsentStatus | ConsentSource MessageDirection | MessageStatus ScheduledMessageStatus | UserRole
BaseEntity Id | TenantId | CreatedAt | UpdatedAt IsDeleted | DeletedAt
FlowPilot.Shared — Cross-Cutting
Result<T> Pattern
ICurrentTenant
IFeatureGate


Triggers
Customer Books (Public Page)
AutoConfirm Worker (3h before)
AutoCompletion Worker (after EndsAt)
Twilio Webhook (Inbound SMS)
Staff Action (Dashboard)
STOP keyword
MediatR Domain Events
AppointmentCreated Event
AppointmentStatus ChangedEvent
InboundSmsReceived Event
CustomerOptedOut Event
Event Handlers
AppointmentBooked SmsHandler instant confirmation SMS
ReminderOptimization Agent schedules 2 reminders
AppointmentRealtime Handler pushes to SignalR
AppointmentStatus ChangedHandler creates audit log
ReviewRecovery Agent sends review request
SmsRealtime Handler pushes to SignalR + ReplyHandlingAgent
CustomerOptedOut Handler cancels pending SMS
agent classifies
Side Effects
ScheduledMessage Created
SMS Sent (Twilio)
SignalR Push (Dashboard)
DB Update (Status / Audit)


Customer Phone
Twilio
FlowPilot API
MessagingService
MediatR
ReplyHandling Agent
Azure OpenAI
PostgreSQL
SignalR
Staff Dashboard
BOOKING FLOW
POST /api/v1/public/book/{slug}
Find-or-create Customer (OptedIn)
Create Appointment (Scheduled)
Publish AppointmentCreatedEvent
par
[Instant Confirmation]
AppointmentBookedSmsHandler
Send SMS (FR/EN)
"Votre RDV est planifie. Repondez OUI!"
[Schedule Reminders]
ReminderOptimizationAgent
ScheduleSms (24h before)
ScheduleSms (3-4h before)
[Real-time Update]
AppointmentRealtimeHandler
New appointment notification
REMINDER DISPATCH (Worker)
ScheduledMessageDispatcher polls
Send reminder SMS
"Rappel: votre RDV demain a 10h. OUI pour confirmer"
CUSTOMER REPLY
"Oui je confirme"
POST /api/webhooks/twilio/incoming
TwilioSignatureFilter (HMAC-SHA1)
ProcessInboundAsync
Save inbound Message
Publish InboundSmsReceivedEvent
par
[Real-time + Agent]
SmsRealtimeHandler
New message in inbox
[Intent Classification]
ReplyHandlingAgent
Classify intent (all upcoming appointments)
Confirm, confidence 0.92
Appointment.Status = Confirmed
Publish AppointmentStatusChangedEvent
AUTO-CONFIRM (if no reply)
AppointmentAutoConfirmWorker
Scheduled -> Confirmed
(3h before StartsAt)
AUTO-COMPLETE (after appointment)
AppointmentAutoCompletionWorker
Confirmed -> Completed
(after EndsAt)


Customer books
(public page or staff)
Scheduled
Triggers on entry:
-
AppointmentBookedSmsHandler
(instant SMS)
- ReminderOptimizationAgent
(schedule reminders)
- AppointmentRealtimeHandler
(SignalR)
Auto-confirm worker
(3h before StartsAt)
Customer replies YES
(ReplyHandlingAgent)
Staff reschedules
(creates new appointment)
Confirmed
Staff cancels
Customer replies CANCEL
(ReplyHandlingAgent)
Staff reschedules
(creates new appointment)
Staff cancels
Auto-completion worker
(after EndsAt)
Triggers on transition:
-
AppointmentStatusChangedHandler
(audit log)
Rescheduled
Cancelled
Completed
Triggers on transition:
-
AppointmentStatusChangedHandler
(audit log)
- ReviewRecoveryAgent (review
request SMS)